

General Outline

Lecture 1:

A short history of GIA theory (Erik -- 20-30 mins)

Lecture 2:

Broad overview of GIA & model components (Glenn – 15-20 min)

Lecture 3:

Earth Rheology & Rotation (Erik – 50 min)

Lecture 4:

GIA modelling & model applications (Glenn – 50 min)

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**Some Problems of
Geodynamics; Being an Essay
to Which the Adams Prize in
the University of Cambridge
was Adjudged in 1911**



A. E. H. 1863-1940 LOVE

IN THE 20TH S OF GIA THEORY

25-1972

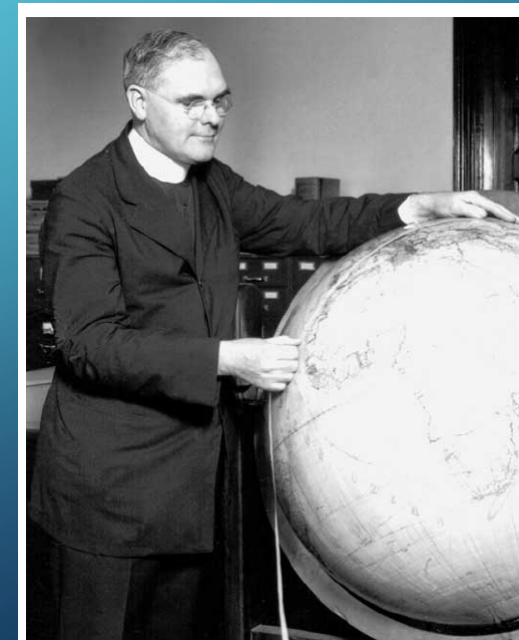
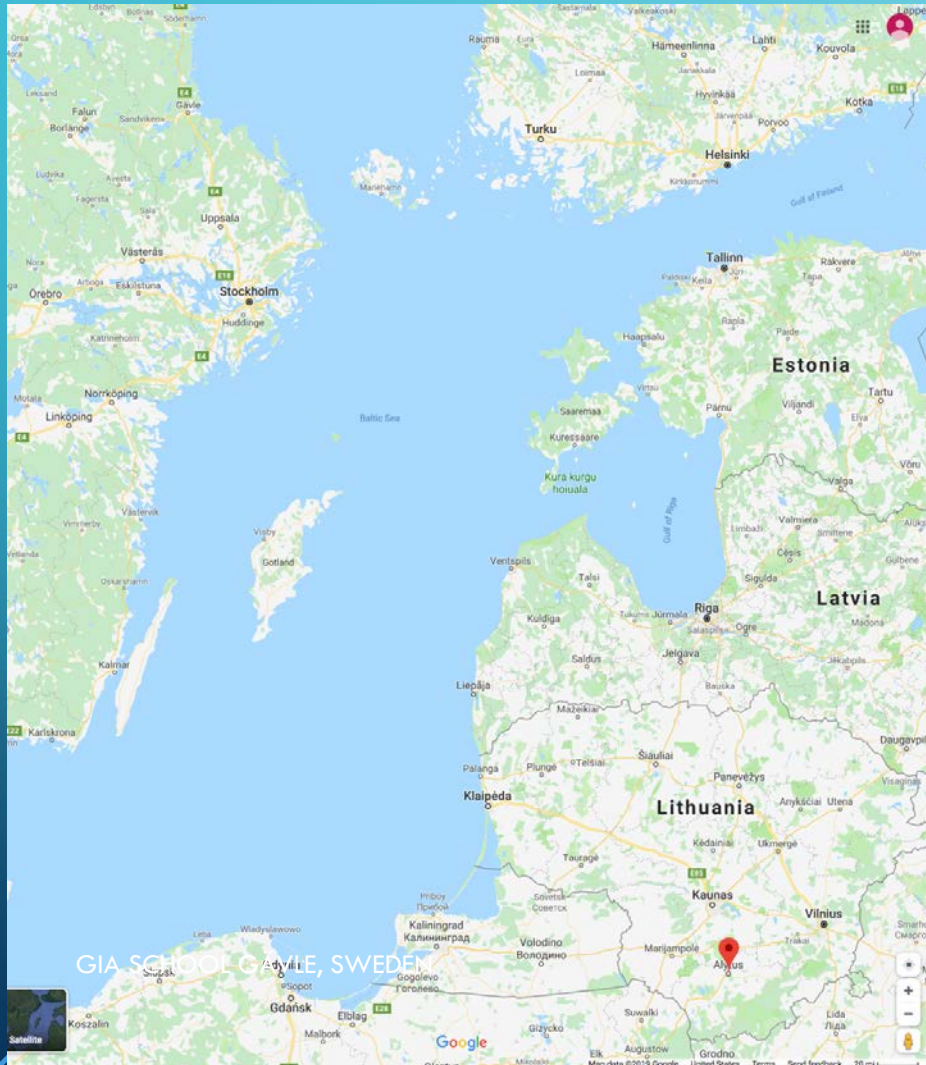


FIGURE 3 James B. Macelwane, S.J.

CHAIM LIEB PEKERIS



Alytus, Lithuania

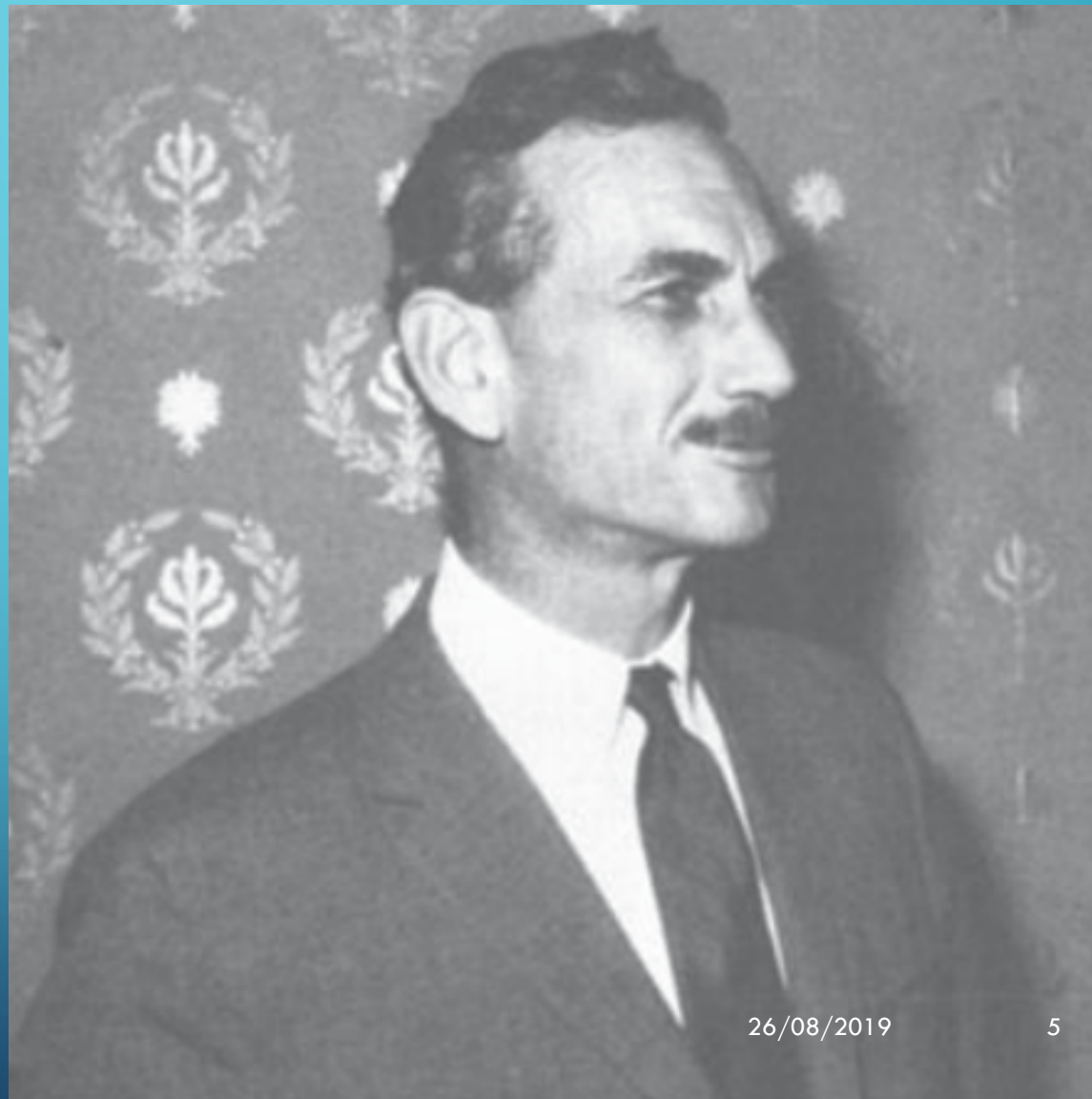
LOUIS B. SLICHTER



NORMAN HASKELL

Mantle Viscosity work at Harvard University:
1935-1937

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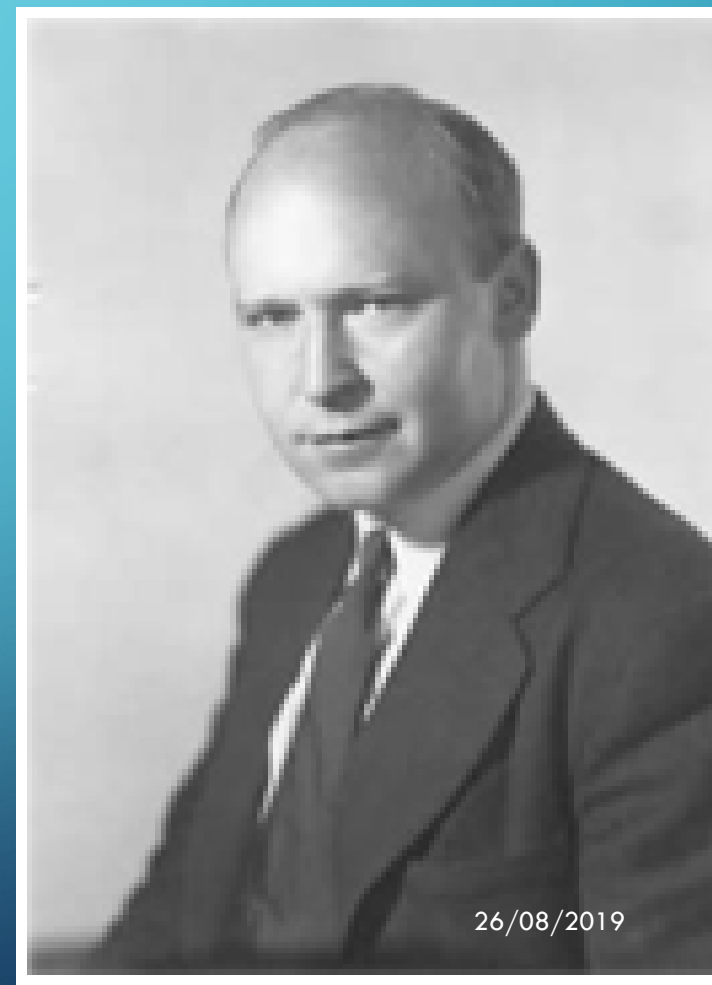
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CARL-GUSTAV ROSSBY



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METEOROLOGY

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WW II (1939-1945)

Meteorology
Ocean science
Seismology

The Weizmann Institute, Rehovot, Israel

University of California, Institute of Geophysics and Planetary Physics

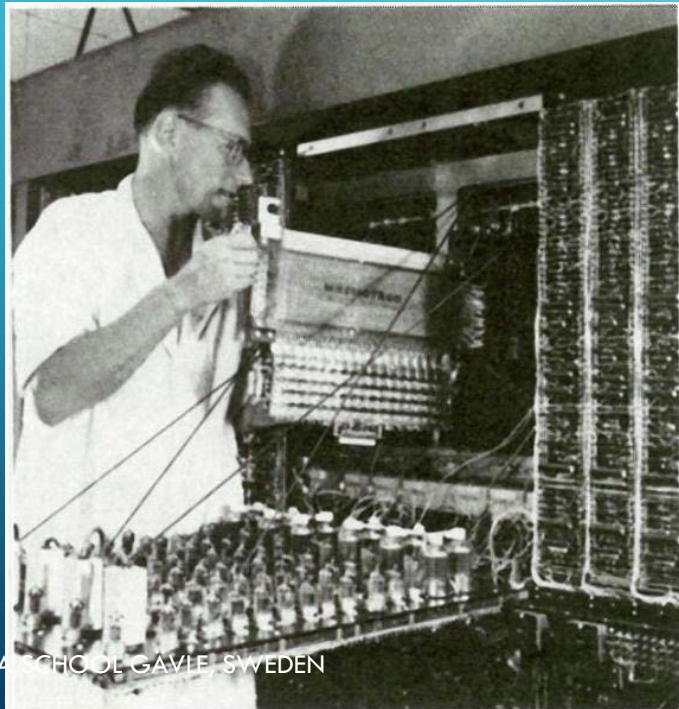
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1946-1960

- Competition for world's top scientists

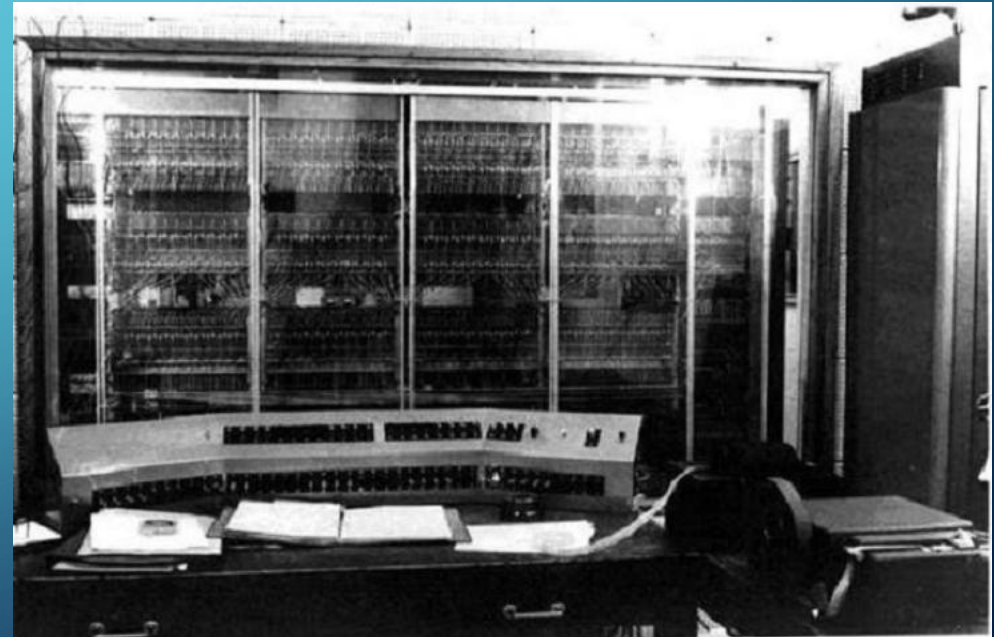


GIA SCHOL GAVLE, SWEDEN

Installation of the Telemeter Magnetic Ferrite Core Memory



2000 heated triodes in the WEIZAC



The console of the WEIZAC

CL PEKERIS: FLUID DYNAMICIST & SEISMOLOGIST



Chaim Pekeris and colleagues discussing model of ocean tides

THE WEIZAC & THE GREAT CHILEAN EARTHQUAKE

Oscillations of the earth*

BY Z. ALTERMAN, H. JABOSCH AND C. L. PEKERIS

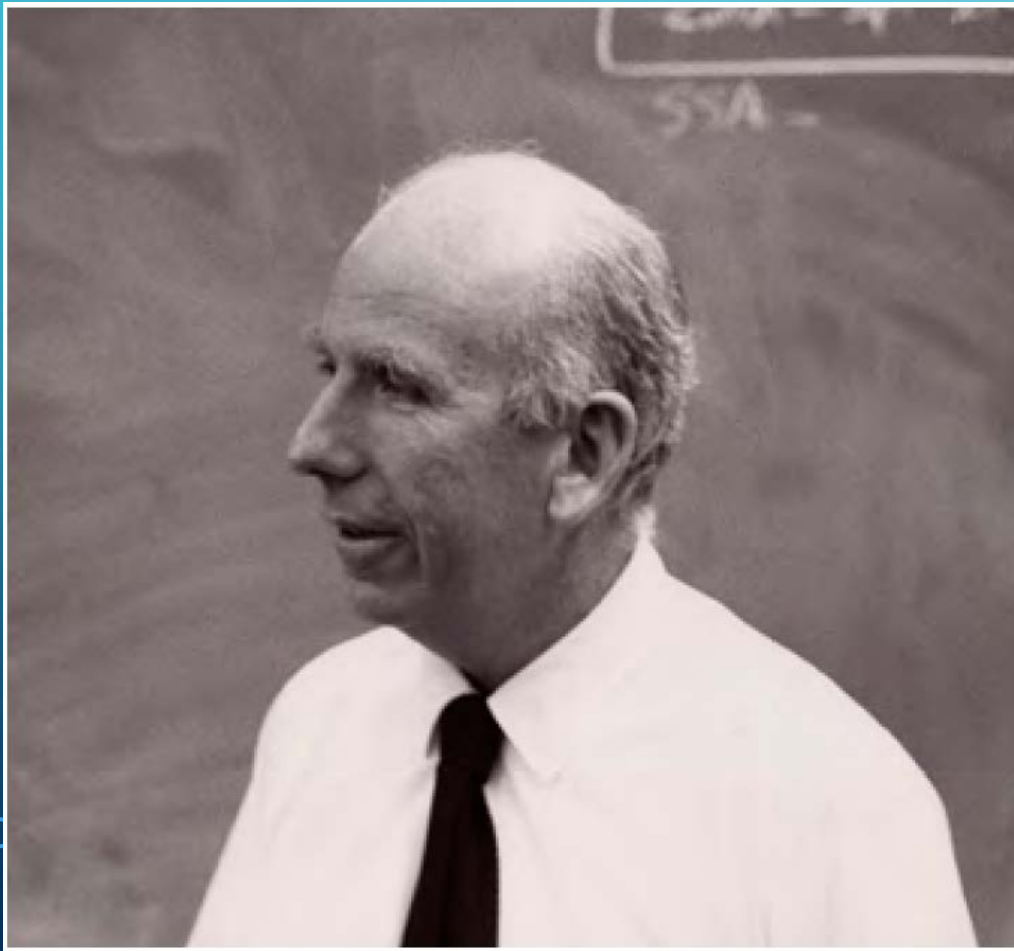
Department of Applied Mathematics, The Weizmann Institute, Rehovot, Israel

(Communicated by Sir Harold Jeffreys, F.R.S.—Received 13 February 1959)

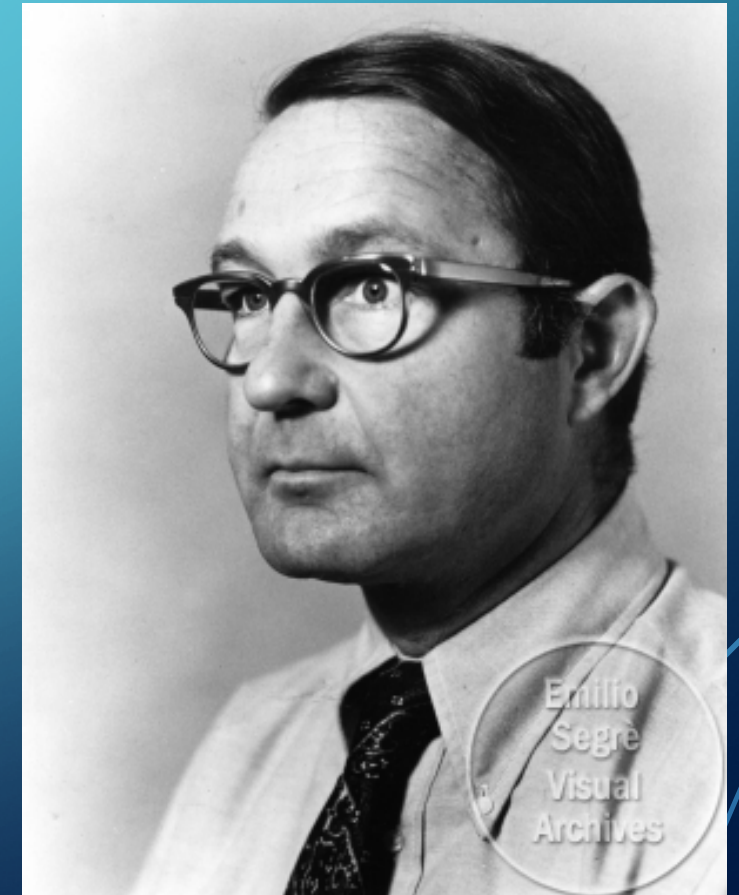


EMERGENCE OF THE MODERN THEORY OF SEISMOLOGY (1955-1965)

Leon Knopoff



Fremann Gilbert



Computation at the Weizmann Institute & by Longman at UCLA (1960-1962)

Goal:

- To compute $\mathbf{u}(r, \theta, \phi)$, $\epsilon(r, \theta, \phi)$, $\sigma(r, \theta, \phi)$ perturbed by applied surface (load or potential) forcing.

$$\sigma(r, \theta, \phi) = 2\mu \epsilon(r, \theta, \phi) + \lambda \operatorname{tr}[\epsilon(r, \theta, \phi)] \mathbf{I}$$

Field equations:

- Perturbation equations of motion and gravitation

$$\begin{aligned} \rho_1^e + \nabla \cdot (\rho_0 \mathbf{u}) &= 0 \\ \rho_0 d^2 \mathbf{u} / dt^2 &= \nabla \cdot \sigma_1^e + \rho_0 \nabla \Phi_1^e + \rho_1^e \nabla \Phi_0 + \rho_0 \nabla V \\ \Delta \Phi_1^e &= -4\pi G \rho_1^e \end{aligned}$$

Solution techniques:

- All variables are decomposed in vector spherical harmonics

$$\mathbf{u}(r, \theta, \varphi) = \sum_{n=0}^{\infty} \sum_{m=0}^n \left(u_{rnm}^{c,s} Y_{nm}^{c,s} \mathbf{e}_r + u_{snm}^{c,s} r \nabla Y_{nm}^{c,s} + u_{tnm}^{c,s} \mathbf{r} \wedge \nabla Y_{nm}^{c,s} \right)$$

$$u_{rn} = y_{1n} Y_n$$

$$\mathbf{u}_{sn} + \mathbf{u}_{tn} = y_{3n} r \nabla Y_n + y_{7n} \mathbf{r} \wedge \nabla Y_n$$

displacement

$$T_{rn} = y_{2n} Y_n$$

$$\mathbf{T}_{sn} + \mathbf{T}_{tn} = y_{4n} r \nabla Y_n + y_{8n} \mathbf{r} \wedge \nabla Y_n$$

traction

$$\Phi_{1n}^e + V_n = y_{5n} Y_n$$

gravitational potential

$$y_{6n} = \dot{y}_{5n} - 4\pi G \rho_0 y_{1n}$$

Lagrangian attraction

- Field equations are reduced to “*y-system of equations*”

$$\begin{aligned} \dot{y}_1 &= \frac{-2\lambda}{\lambda+2\mu} \frac{y_1}{r} + \frac{1}{\lambda+2\mu} y_2 + \frac{\lambda n(n+1)}{\lambda+2\mu} \frac{y_3}{r} \\ \dot{y}_2 &= \left[-4\rho_0 g_0 r + \frac{4\mu(3\lambda+2\mu)}{\lambda+2\mu} \right] \frac{y_1}{r^2} - \frac{4\mu}{\lambda+2\mu} \frac{y_2}{r} + n(n+1) \left[\rho_0 g_0 r - \frac{2\mu(3\lambda+2\mu)}{\lambda+2\mu} \right] \frac{y_3}{r^2} + n(n+1) \frac{y_4}{r} - \rho_0 y_6 \\ \dot{y}_3 &= -\frac{y_1}{r} + \frac{y_3}{r} + \frac{y_4}{\mu} \\ \dot{y}_4 &= \left[\rho_0 g_0 r - \frac{2\mu(3\lambda+2\mu)}{\lambda+2\mu} \right] \frac{y_1}{r^2} - \frac{\lambda}{\lambda+2\mu} \frac{y_2}{r} + \frac{2\mu}{\lambda+2\mu} [\lambda(2n^2+2n-1) + 2\mu(n^2+n-1)] \frac{y_3}{r^2} - \frac{3}{r} y_4 - \frac{\rho_0}{r} y_5 \\ \dot{y}_5 &= -4\pi G \rho_0 y_1 + y_6 \\ \dot{y}_6 &= -4\pi G \rho_0 n(n+1) \frac{y_3}{r} + n(n+1) \frac{y_5}{r^2} - 2 \frac{y_6}{r} \end{aligned}$$

-- displacement & attraction vanish at Earth's center-of-mass

-- y_i are continuous across interfaces

-- for surface load: $y_2(a) = -P_n$ and $y_4(a) = 0$

-- $y_6(a) + \frac{n+1}{a} y_5(a) = \frac{2n+1}{a} V_n$

Summary:

- Given $\rho(r)$, $\lambda(r)$, $\mu(r)$, solve the BVP for $y_i(r)$

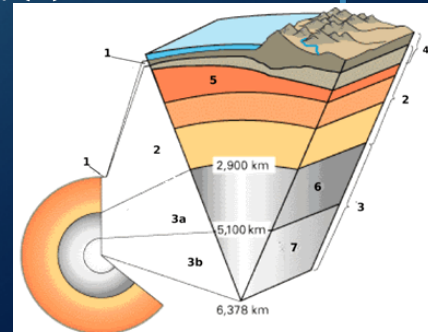
- Retrieve Green's kernels

$$\mathcal{G}_1(\alpha, r) = \sum_{n=0}^{\infty} y_{2n}(r) \mathcal{P}_n(\cos \alpha)$$

- Convolve with loads:

$$u_1(r, \theta, \phi) = \int \mathcal{G}_1 L(\theta, \phi) dS$$

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MODERN GIA THEORY AS SEISMOLOGICAL ANALOGUE

VOL. 12, NO. 4

REVIEWS OF GEOPHYSICS AND SPACE PHYSICS

NOVEMBER 1974

The Impulse Response of a Maxwell Earth

W. R. PELTIER¹

University of Colorado-NOAA Cooperative Institute for Research in Environmental Sciences, Boulder, Colorado 80302

40 YEARS EARLIER AT MIT WITH SLICHTER

AUGUST, 1935

PHYSICS

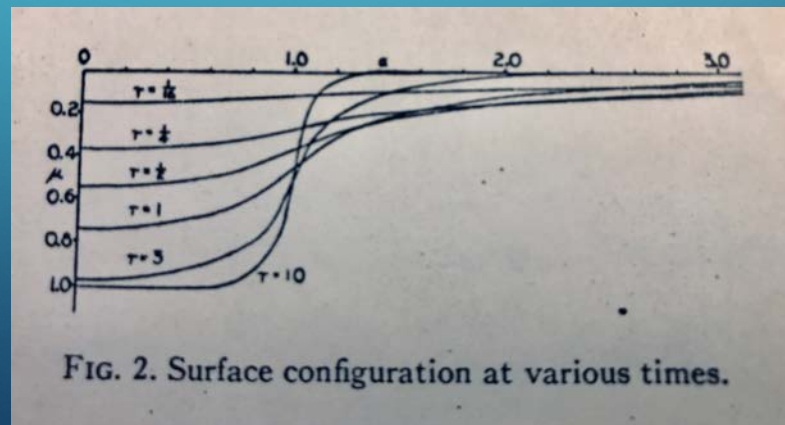
VOLUME 6

The Motion of a Viscous Fluid Under a Surface Load

N. A. HASKELL, *Massachusetts Institute of Technology*

(Received April 25, 1935)

A formal solution is given for the motion of a highly viscous fluid when a symmetrical pressure is applied at the surface. This is applied to the subsidence of a cylindrical body of constant thickness and to the recovery of the fluid after removal of a load. Applying the latter case to the plastic recoil of the earth after the disappearance of the Pleistocene ice sheets, it is found that the geological data imply a kinematic viscosity of the order of 3×10^{21} c.g.s. units.



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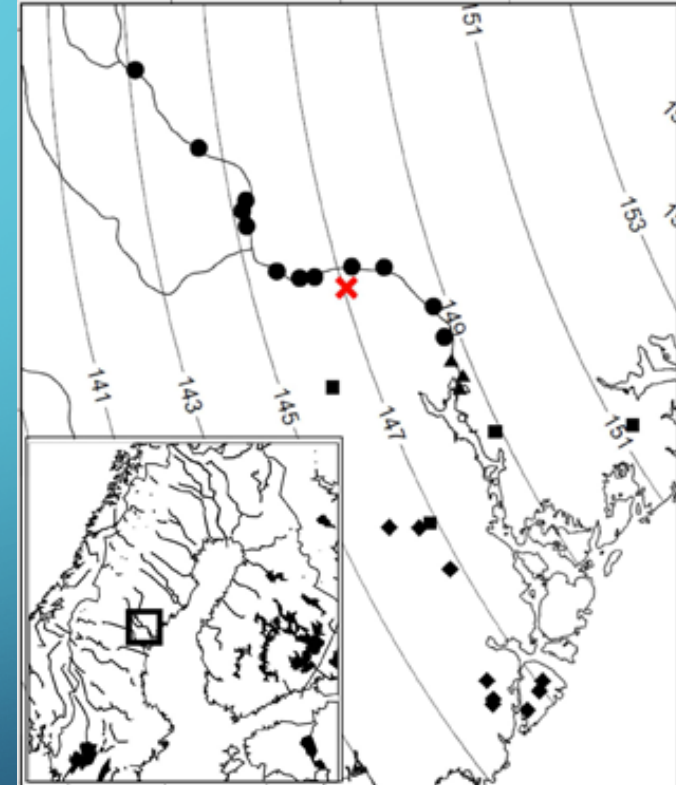
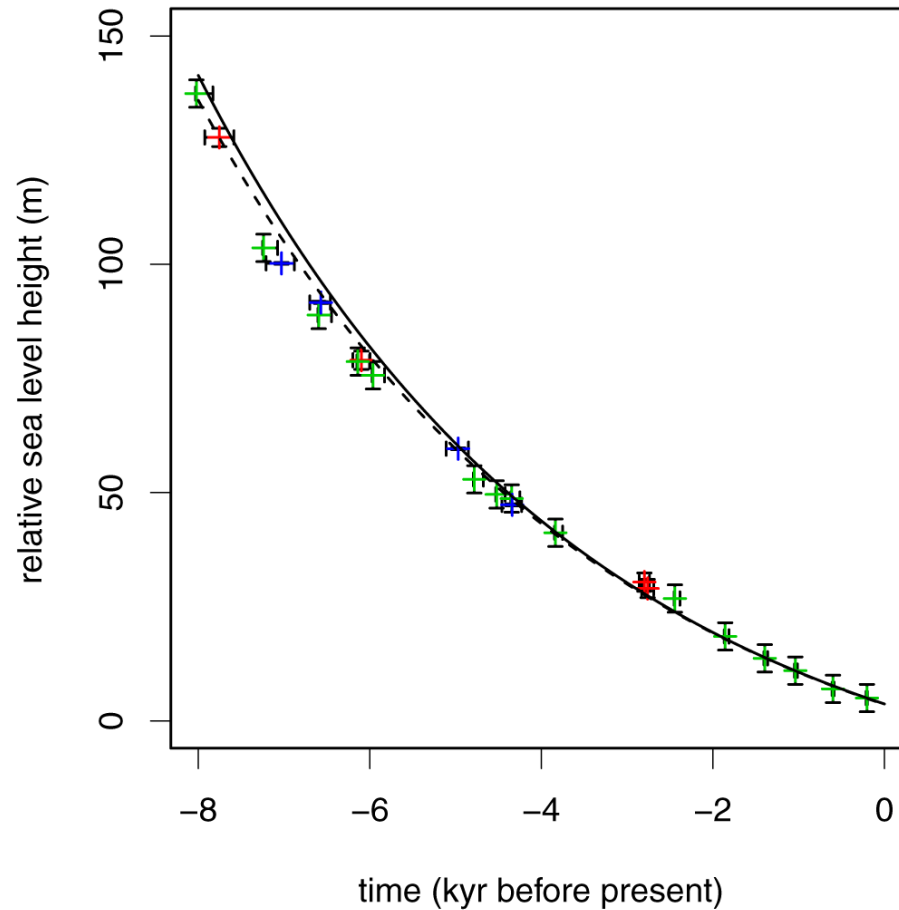
Reappraisal of the Ångerman River decay time estimate and its application to determine uncertainty in Earth viscosity structure

Maaria Nordman,^{1,2} Glenn Milne² and Lev Tarasov³

¹Finnish Geospatial Research Institute, National Land Survey of Finland, Kirkkonummi, Finland. E-mail: maaria.nordman@nls.fi

²Department of Earth Sciences, University of Ottawa, Ottawa, Canada

³Department of Physics and Physical Oceanography, Memorial University, St. John's, Newfoundland, Canada



Ångerman River emergence data fit to exponential curves: $E_h = A (e^{-t/\tau} - 1) + c$,

EXPONENTIAL RSL CURVE RESULT

- $0.3 \times 10^{21} \leq \eta_{UM} \leq 1.0 \times 10^{21} \text{ Pa sec}$ (upper mantle)
- diminished constraint of viscosity in the top half of the lower mantle: $1.0 \times 10^{21} \leq \eta_{UM} \leq 90 \times 10^{21} \text{ Pa sec}$

$$(1 \text{ Pa} = 1 \text{ kg} / \text{m sec}^2)$$

END OF LECTURE 1 (ERIK IVINS)